

R-Super MedFormer on the PanTS test set

Evaluation with SegEvalKit 0.1.0 (2026-09-30): 901 CTs, 17 structures, organ, tumour, patient and cohort level. All numbers are generated from the evaluation outputs.

1 Setup

Model	MedFormer, checkpoint AbdomenAtlas/MedFormerPanTS (file fold_0_latest.pth): the R-Super repository’s “Baseline (Demo, no report supervision)”, trained with PanTS masks only. The report-supervised R-SuperPanTSMerlin was not run.
Code	Official MrGiovanni/R-Super, commit fa8fdce (2026-09-24), predict_abdomenatlas.py with --organ_mask_on_lesion, --save_probabilities_lesions and --organs_with_tumor pancreas; A100 GPUs via SLURM.
Fix	Upstream pad_to_training_size pads the wrong axis for scans with fewer than 128 voxels along z or x after 1 mm resampling, and 25 test scans failed to save. We ran a copy with the two pad calls swapped (benchmarks/pants/patch_medformer.py); all other scans are unaffected.
Data	PanTS official test split: 901 CTs, 151 with a pancreatic tumour and 750 without; per-structure reference masks; metadata (phase, scanner, country, slice thickness, sex, age).
Protocol	Pancreas scored as pancreas $\cup$ lesion (the lesion mask is not always inside the pancreas mask). segevalkit audit: in 227 scans a label header disagrees with its CT (218 aorta files) while the voxels are aligned, so masks are compared voxel by voxel. NSD tolerance 2 mm; tumour blobs under 10 voxels ignored; empty vs. non-empty masks score 0 (overlap) and the image diagonal (distances).
Comparator	TotalSegmentator v2 (total task) on the same scans.

2 Reported vs. reproduced

R-Super reports patient-level tumour detection on the PanTS test set: a patient is called positive when the predicted tumour volume exceeds a threshold. We reproduce this twice (Table 1): **(A)** with their own code path, i.e. their detection() volume (1 mm resampling and a 1-erode / 2-dilate opening), their ground truth (number of pancreatic lesion instances ≥ 1 in metadata_pants.csv) and their threshold grid; and **(B)** with the raw predicted volume and the reference masks of SegEvalKit. In both we pick the threshold whose specificity is closest to their 91%; brackets are 95% bootstrap intervals over patients, with the threshold re-chosen in each resample. The two ground truths agree on 901 of 901 patients.

Table 1: Patient-level detection on PanTS test (%). $F1 = 2TP / (2TP + FP + FN)$ over patients.

Patient level, PanTS test (%)	Sensitivity	Specificity	F1	AUC
R-Super GitHub, this checkpoint ^a	76	91	69	–
R-Super protocol (their volume, their thresholds)	77 [68, 85]	91	69 [63, 74]	0.907
SegEvalKit protocol (raw predicted volume)	77 [68, 84]	91	69 [63, 74]	0.905
R-Super GitHub, R-Super model (not run)	80	93	74	–

^aRead from the “PanTS test” panel of documents/demo_results.png, linked from rsuper_train/Merlin_demo.md of the R-Super repository (grey bars, “standard segmentation, trained w/ 0.9K pancreatic tumor masks”); the threshold used there is not stated.

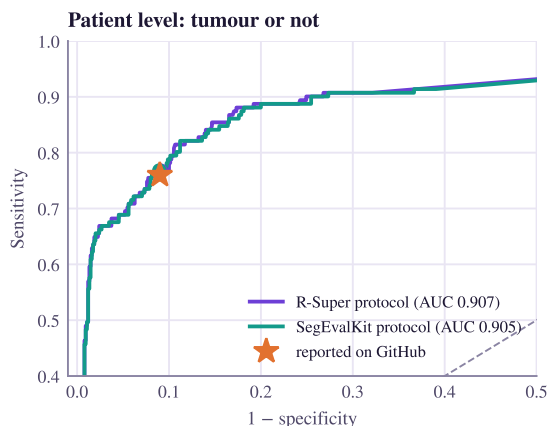


Figure 1: ROC of both protocols; dots: our operating points; star: the reported point.

The reported numbers reproduce. With their protocol (threshold 1.10 mL) we obtain sensitivity 77% [68, 85] at specificity 91% and F1 69% [63, 74], against the reported 76 / 91 / 69; the reported sensitivity is within and the reported F1 within our interval. The AUC is 0.907. With SegEvalKit’s raw volume the curve is similar (AUC 0.905; at specificity ≥ 0.90 , sensitivity 0.788). Remaining differences: the unknown threshold of the published figure, 25 scans that the unpatched code cannot process (2 of them with a tumour), and our single inference run.

3 Where it works and where it fails

3.1 Organs: accurate in most patients, with a heavy HD95 tail

Table 2: MedFormer per structure (901 cases; the lesion row includes tumour-free patients). TS Dice: mean Dice of TotalSegmentator; MF better: fraction of patients in which MedFormer’s Dice is higher (paired, same cases).

Structure	Dice mean / median	NSD mean / median	HD95 (mm) mean / median	TS Dice	MF better
Pancreas	.864 / .910	.854 / .911	19.5 / 2.9	.823	91%
Pancreatic lesion	.597 / 1.000	.590 / 1.000	206.5 / 0.0	–	–
Liver	.971 / .981	.929 / .961	7.8 / 1.9	.963	93%
Spleen	.928 / .972	.923 / .988	31.7 / 1.5	.946	79%
Kidney (L)	.889 / .966	.880 / .970	45.4 / 1.8	.890	85%
Kidney (R)	.939 / .969	.928 / .972	22.6 / 1.7	.905	90%
Stomach	.903 / .955	.857 / .925	16.9 / 2.8	.871	90%
Gallbladder	.805 / .913	.823 / .935	56.7 / 2.6	.755	84%
Duodenum	.897 / .925	.910 / .942	15.6 / 2.5	.737	98%
Colon	.921 / .942	.899 / .927	12.3 / 3.3	–	–
Adrenal (L)	.837 / .877	.928 / .982	54.4 / 1.5	–	–
Adrenal (R)	.843 / .874	.940 / .981	60.2 / 1.5	–	–
Aorta	.903 / .936	.914 / .948	15.0 / 2.7	.881	80%
IVC	.854 / .913	.853 / .901	22.0 / 4.3	.817	88%
SMA	.836 / .863	.947 / .982	32.4 / 1.5	–	–
Veins	.860 / .890	.922 / .949	21.4 / 2.5	–	–
Common bile duct	.643 / .730	.768 / .889	128.4 / 5.8	–	–

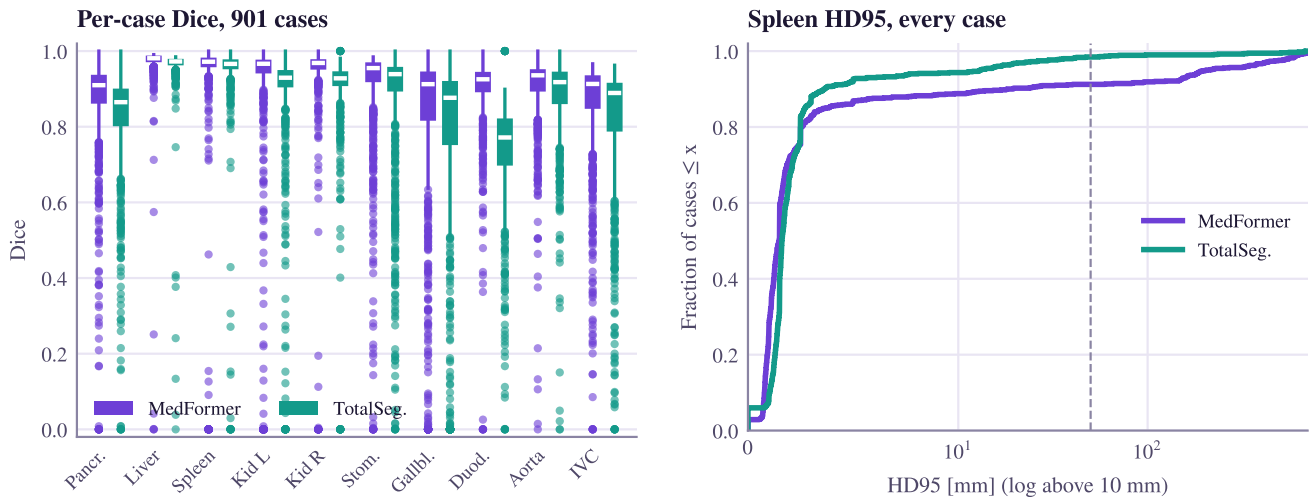


Figure 2: Left: per-case Dice of both models on ten shared organs. **Right:** cumulative spleen HD95; about one MedFormer case in ten lies above 50 mm (dashed).

Good. Mean Dice is ≥ 0.85 on 11 of 16 organs and vessels (pancreas .864, median .910; liver .971). Paired with TotalSegmentator on the same patients, MedFormer has the higher Dice in 79–98% of patients on every shared organ, largest on the duodenum. **Weak.** The common bile duct (.643) and the gallbladder (.805 macro, .879 pooled) are the weakest structures. Mean HD95 is far above the median (spleen 31.7 vs. 1.5 mm): the 79 spleen cases above 50 mm make up 94% of the mean. 29 of them are scans with an *empty* reference spleen mask on which MedFormer still predicts a spleen (55 such scans; TotalSegmentator predicts one in 1), and the other 50 are good segmentations (median Dice .957) with stray voxels far from the organ. Excluding empty references, mean spleen HD95 is 14.2 mm. The left kidney shows the same pattern (101 cases, 47 with an empty reference). This tail is why MedFormer’s mean HD95 is worse than TotalSegmentator’s on 7 of ten organs although its HD95 is lower in most patients on 10 of them, and why the two ranking schemes disagree on the spleen and the kidney (left): ranking by mean Dice puts TotalSegmentator first, ranking per patient puts MedFormer first (bootstrap Kendall’s $\tau \geq 1.00$ for both schemes).

3.2 Tumours: large ones found, small ones missed, false alarms in tumour-free patients

Table 3: Pancreatic tumours (161 reference tumours in 151 patients).

Tumour metric (95% CI)	MedFormer
Tumours found (of 161)	0.81 [0.75, 0.88]
Predicted tumours that are real	0.24 [0.20, 0.27]
False tumours per scan	0.49 [0.44, 0.55]
Pooled tumour Dice	0.55 [0.47, 0.62]
False tumours per scan: tumour-free / tumour pat.	0.52 / 0.34
Per tumour patient: lesion precision	.821
Per tumour patient: tumour Dice (mean / median)	.536 / .681
Lesion probabilities: ECE / Brier	0.040 / 0.043
FROC CPM (1/8–8 FP per scan)	0.803 [0.73, 0.87]
Lesion-level average precision	0.599
Patient sens. at spec. 0.90, plain / localized	0.788 / 0.762
Patient AUC, plain / localized	0.905 / 0.816

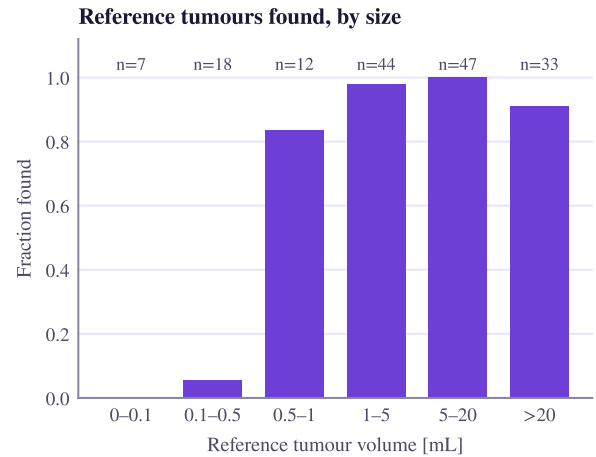


Figure 3: Fraction of reference tumours found, by volume.

MedFormer finds 131 of 161 tumours: 97% of those of 1 mL or more, but only 4% of the 25 below 0.5 mL. Only 0.24 of its predicted tumours are real (0.49 false tumours per scan); most false alarms are in tumour-free patients (0.52 per scan vs. 0.34 in tumour patients) and are small (median 0.26 mL), so in tumour patients the per-patient lesion precision is .821. When a tumour is present, its voxel Dice is .536 on average (median .681). The lesion probabilities are well calibrated in the band around the masks.

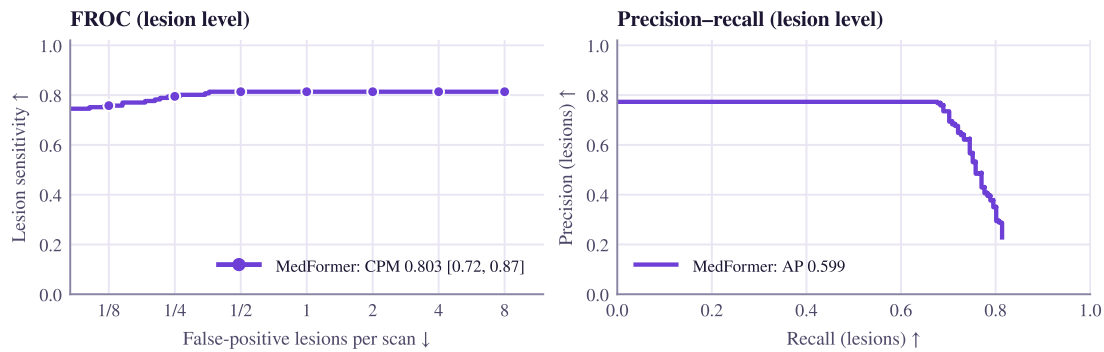


Figure 4: Lesion-level detection curves, ranking each predicted tumour by its maximum probability: FROC with the CPM (left) and precision-recall with the average precision (right), over all 161 reference tumours.

The FROC score (CPM) is 0.803: sensitivity is 0.76 at 1/8 and 0.81 at one false tumour per scan, and flat beyond, because the tumours still missed are never predicted at any threshold; the lesion-level average precision is 0.60. *Localized* patient-level detection counts a flagged patient only if a predicted tumour overlaps the reference tumour: it lowers the sensitivity at specificity 0.90 from 0.788 to 0.762 and the AUC from 0.905 to 0.816, so some patients are flagged for a false tumour in the wrong place.

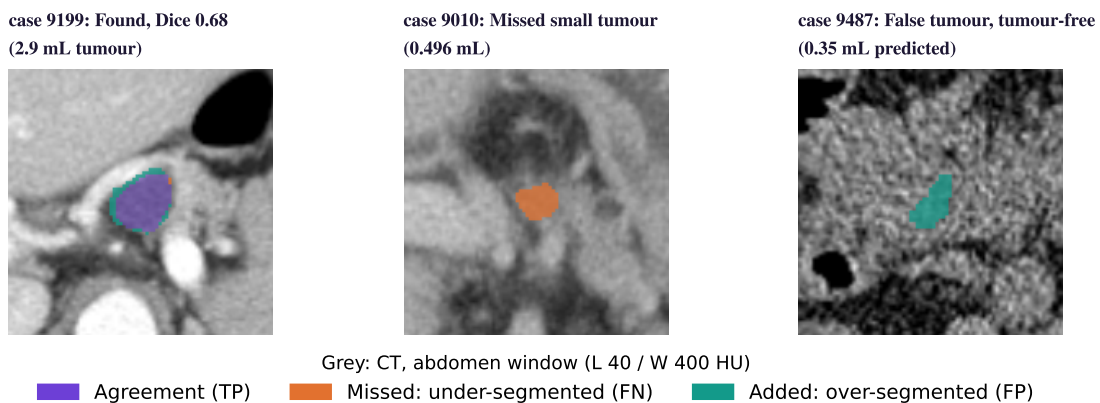


Figure 5: Three real test cases, chosen by rule: the found tumour whose Dice is closest to the per-patient median, the largest missed tumour under 0.5 mL, and the tumour-free patient whose false-tumour volume is the median.

Table 4: By patient group (mean; Kruskal–Wallis or Mann–Whitney test, Holm-corrected over all 63 cohort tests).

Factor	group (<i>n</i>)	Pancreas Dice	Pancreas NSD	Lesion Dice
CT phase	Arterial (162)	.838	.832	.612
	Delay (12)	.875	.847	.590
	Non-contrast (134)	.806	.776	.599
	Venous (593)	.884	.878	.593
	<i>adjusted p</i>	$< 10^{-3}$	$< 10^{-3}$	1.000
Scanner	GE (42)	.719	.695	.517
	Philips (11)	.819	.752	.556
	Siemens (334)	.850	.830	.590
	<i>adjusted p</i>	$< 10^{-3}$	0.004	1.000
Slice thickness	≤ 1.25 mm (246)	.793	.790	.669
	1.25–3 mm (461)	.890	.869	.602
	> 3 mm (194)	.894	.900	.495
	<i>adjusted p</i>	$< 10^{-3}$	$< 10^{-3}$	0.005

3.3 Cohorts: contrast phase, scanner and slice thickness matter

Pancreas Dice is .884 on venous-phase and .806 on non-contrast scans, and .850 on Siemens against .719 on GE scanners. Thin slices score lower (.793) than thick slices (.894), the opposite of a resolution effect, which points to confounding by site and case mix; the groups are observational. 25 of 63 cohort tests are significant.

4 Metric definitions

P prediction, G reference, $TP/FP/FN$ voxel counts; $D_{P \rightarrow G}$ distances (mm) from each surface voxel of P to the surface of G ; ∂P surface of P ; “les” marks counts of tumour blobs (a reference tumour is found when a predicted blob overlaps it).

Metric	Formula	Meaning	Better
Dice	$2TP/(2TP + FP + FN)$	Overlap of prediction and reference	↑
NSD $_{\tau}$	$(D_{P \rightarrow G} \leq \tau + D_{G \rightarrow P} \leq \tau)/(\partial P + \partial G)$	Fraction of both surfaces within $\tau = 2$ mm of the other	↑
HD95	$\max(P_{95}(D_{P \rightarrow G}), P_{95}(D_{G \rightarrow P}))$	Boundary error that 95% of surface points stay within	↓
Pooled Dice	$2 \sum_i TP_i / (2 \sum_i TP_i + \sum_i FP_i + \sum_i FN_i)$	Dice over all voxels of all cases (large structures weigh more)	↑
Tumours found	$TP_{\text{les}} / (TP_{\text{les}} + FN_{\text{les}})$	Fraction of reference tumours overlapped by a prediction	↑
Predicted tumours real	$TP_{\text{les}} / (TP_{\text{les}} + FP_{\text{les}})$	Fraction of predicted tumours that overlap a reference tumour	↑
False tumours / scan	$\sum FP_{\text{les}} / n_{\text{scans}}$	Predicted tumours overlapping no reference, per scan	↓
Patient sens. / spec.	$TP/(TP + FN), TN/(TN + FP)$ over patients	Tumour patients flagged / tumour-free patients not flagged	↑
Patient F1	$2TP/(2TP + FP + FN)$ over patients	Balance of patient-level sensitivity and precision	↑
AUC	$\int \text{TPR} d\text{FPR}$	Probability that a tumour patient gets a larger predicted volume than a tumour-free one	↑
Localized sensitivity	$TP_{\text{loc}} / (TP + FN)$ over patients	A flagged tumour patient counts only if a predicted tumour overlaps the reference tumour	↑
CPM (FROC)	$\frac{1}{7} \sum_{r \in \{\frac{1}{8}, \dots, 8\}} \text{sens}(r)$	Mean tumour sensitivity at 1/8 to 8 false tumours per scan (LUNA16)	↑
Lesion AP	$\sum_k (R_k - R_{k-1}) P_k$	Area under the tumour-level precision–recall curve, tumours ranked by probability	↑
ECE	$\sum_b \frac{ B_b }{N} \text{acc}(B_b) - \text{conf}(B_b) $	Gap between confidence and accuracy (15 bins)	↓
Brier	$\frac{1}{N} \sum_i (p_i - g_i)^2$	Mean squared error of the voxel probabilities	↓

Statistics: 95% percentile bootstrap intervals over patients; paired Wilcoxon tests with Holm correction (all 30 of 30 MedFormer vs. TotalSegmentator comparisons significant); ranking by mean score (aggregate-then-rank) or by mean per-patient rank (rank-then-aggregate).

5 Limitations and reproduction

- PanTS publishes no evaluation script for organ metrics; NSD tolerance, detection rule and empty-mask policy are SegEvalKit’s documented defaults, so organ and lesion-level metrics are not leaderboard numbers; the patient-level comparison of Section 2 uses R-Super’s own code, matched on specificity because the published threshold is not stated.
- Some reference masks are empty for organs present in the scan field (e.g. 55 spleens); a prediction there counts as a large error.

- Only the mask-trained checkpoint was evaluated, with one disclosed fix to the official inference code.

```
sbatch benchmarks/pants/run_medformer.sbatch slurm/pants_eval.sbatch slurm/lesion_scores.sbatch
python benchmarks/pants/{detection_figures,rsuper_detection}.py reports/rsuper_medformer/make_report.py
```